

Gaussian kicks amid coherent evolution: ensemble-averaged error of the joint and split propagators

Nathaniel Leitaο

October 1, 2026

Abstract

We take the simplest stochastic Trotter model: one step is the unitary $\exp(i \sum_{\alpha} X_{\alpha} O_{\alpha})$ with jointly Gaussian X_{α} of mean $\mu_{\alpha} \delta t$ and covariance $\Gamma_{\alpha\beta} \delta t$. We compute, to leading order in δt , (i) the ensemble-averaged error of the joint (unsplit) channel relative to the target Lindbladian $\mathcal{L} = i \sum_{\alpha} \mu_{\alpha} \text{ad}_{O_{\alpha}} - \frac{1}{2} \sum_{\alpha\beta} \Gamma_{\alpha\beta} \text{ad}_{O_{\alpha}} \text{ad}_{O_{\beta}}$, and (ii) the extra error from splitting the propagator into the product $\prod_{\alpha} \exp(i X_{\alpha} O_{\alpha})$. The joint channel errs at $\mathcal{O}(\delta t^2)$ per step. The split channel generally errs at $\mathcal{O}(\delta t)$ per step: the off-diagonal covariance on non-commuting terms becomes a coherent Hamiltonian $H_{\text{shift}} = \frac{i}{2} \sum_{\alpha < \beta} \Gamma_{\alpha\beta} [O_{\alpha}, O_{\beta}]$, so the averaged split circuit converges to the wrong generator. For diagonal Γ the split channel is exactly the Lie–Trotter splitting of $\mathcal{L}$. A palindromic ordering of the same sample cancels the $\mathcal{O}(\delta t)$ term. Every formula is checked against exact Gauss–Hermite averages.

1 Setup

Let $O_1, \dots, O_{N_c}$ be Hermitian operators, and write $a_{\alpha} \equiv \text{ad}_{O_{\alpha}} = [O_{\alpha}, \cdot]$. One step uses the random unitary

$$U(X) = \exp\left(i \sum_{\alpha} X_{\alpha} O_{\alpha}\right), \quad X = \mu \delta t + \xi, \quad \xi \sim \mathcal{N}(0, \Gamma \delta t), \quad (1)$$

with $\Gamma = \Gamma^T \succeq 0$. The fluctuations are therefore $\mathcal{O}(\delta t^{1/2})$. With $\text{Ad}_U \rho = U \rho U^{\dagger} = e^{iX \cdot a} \rho$ and $X \cdot a \equiv \sum_{\alpha} X_{\alpha} a_{\alpha}$, the ensemble-averaged joint channel is

$$\Phi = \mathbb{E}[e^{iX \cdot a}]. \quad (2)$$

(With the convention $e^{-i \sum X O}$, flip the sign of μ throughout.) Define

$$M = \sum_{\alpha} \mu_{\alpha} a_{\alpha}, \quad D = \sum_{\alpha\beta} \Gamma_{\alpha\beta} a_{\alpha} a_{\beta}, \quad \mathcal{L} = iM - \frac{1}{2}D, \quad (3)$$

so that $\mathcal{L} \rho = i \sum_{\alpha} \mu_{\alpha} [O_{\alpha}, \rho] - \frac{1}{2} \sum_{\alpha\beta} \Gamma_{\alpha\beta} [O_{\alpha}, [O_{\beta}, \rho]]$ is the target generator. The first-order split channel, with the factors in a fixed order $1, \dots, N_c$ and all driven by the same sample X , is

$$\Phi_{\text{split}} = \mathbb{E}\left[\prod_{\alpha=1}^{N_c} e^{iX_{\alpha} a_{\alpha}}\right] = \mathbb{E}[e^{iX_1 a_1} e^{iX_2 a_2} \dots e^{iX_{N_c} a_{N_c}}], \quad (4)$$

which is the average of $\text{Ad}_{U_1 U_2 \dots U_{N_c}}$.

Moments. To $\mathcal{O}(\delta t^2)$ we need

$$\mathbb{E}[X_\alpha] = \mu_\alpha \delta t, \quad \mathbb{E}[X_\alpha X_\beta] = \Gamma_{\alpha\beta} \delta t + \mu_\alpha \mu_\beta \delta t^2, \quad (5)$$

$$\mathbb{E}[X_\alpha X_\beta X_\gamma] = \delta t^2 S_{\alpha\beta\gamma} + \mathcal{O}(\delta t^3), \quad \mathbb{E}[X_\alpha X_\beta X_\gamma X_\delta] = \delta t^2 T_{\alpha\beta\gamma\delta} + \mathcal{O}(\delta t^3), \quad (6)$$

where

$$S_{\alpha\beta\gamma} = \mu_\alpha \Gamma_{\beta\gamma} + \mu_\beta \Gamma_{\alpha\gamma} + \mu_\gamma \Gamma_{\alpha\beta}, \quad T_{\alpha\beta\gamma\delta} = \Gamma_{\alpha\beta} \Gamma_{\gamma\delta} + \Gamma_{\alpha\gamma} \Gamma_{\beta\delta} + \Gamma_{\alpha\delta} \Gamma_{\beta\gamma} \quad (7)$$

(T is the Isserlis/Wick tensor). All moments of order $n \geq 5$ are $\mathcal{O}(\delta t^3)$.

2 Joint (unsplit) channel versus $e^{\delta t \mathcal{L}}$

Proposition 1.

$$\Phi = e^{\delta t \mathcal{L}} + \delta t^2 E_1 + \mathcal{O}(\delta t^3), \quad E_1 = \frac{i}{12} \sum_{\alpha\beta} \Gamma_{\alpha\beta} [a_\alpha, [a_\beta, M]] + \frac{1}{24} (Q + R - 2D^2), \quad (8)$$

with

$$Q = \sum \Gamma_{\alpha\gamma} \Gamma_{\beta\delta} a_\alpha a_\beta a_\gamma a_\delta, \quad R = \sum_{\alpha\delta} \Gamma_{\alpha\delta} a_\alpha D a_\delta. \quad (9)$$

Proof. Write $K = iX \cdot a = i\delta t M + iN$ with $N = \xi \cdot a$, and expand $\Phi = \sum_n \mathbb{E}[K^n]/n!$.

- $\mathcal{O}(\delta t)$: $\mathbb{E}[K] + \frac{1}{2}\mathbb{E}[K^2] \supset i\delta t M - \frac{1}{2}\delta t D = \delta t \mathcal{L}$. The joint channel is consistent with $\mathcal{L}$.
- $\mathcal{O}(\delta t^2)$, $\mu\mu$: $-\frac{1}{2}\delta t^2 M^2$. This matches $\frac{1}{2}\delta t^2 \mathcal{L}^2 \supset -\frac{1}{2}\delta t^2 M^2$.
- $\mathcal{O}(\delta t^2)$, $\mu\Gamma$: $\frac{1}{6}\mathbb{E}[K^3] \supset \frac{-i}{6}\mathbb{E}[MNN + NMN + NNM] = -\frac{i\delta t^2}{6}(MD + C + DM)$, where $C = \sum \Gamma_{\alpha\beta} a_\alpha M a_\beta$. The target has $\frac{1}{2}\delta t^2 \mathcal{L}^2 \supset -\frac{i\delta t^2}{4}(MD + DM)$. The difference is $\frac{i\delta t^2}{12}(MD + DM - 2C)$. Because Γ is symmetric, $MD + DM - 2C = \sum \Gamma_{\alpha\beta} [a_\alpha, [a_\beta, M]]$.
- $\mathcal{O}(\delta t^2)$, $\Gamma\Gamma$: $\frac{1}{24}\mathbb{E}[N^4] = \frac{\delta t^2}{24} \sum T_{\alpha\beta\gamma\delta} a_\alpha a_\beta a_\gamma a_\delta = \frac{\delta t^2}{24}(D^2 + Q + R)$. The target has $\frac{1}{2}\delta t^2 \mathcal{L}^2 \supset \frac{\delta t^2}{8} D^2 = \frac{3\delta t^2}{24} D^2$. $\square$

The quartic defect can also be written as a commutator:

$$Q + R - 2D^2 = \sum_{\alpha\beta\gamma\delta} \Gamma_{\alpha\beta} \Gamma_{\gamma\delta} a_\alpha \left([a_\gamma, a_\beta] a_\delta + [a_\gamma a_\delta, a_\beta] \right), \quad (10)$$

using $Q - D^2 = \sum \Gamma_{\alpha\beta} \Gamma_{\gamma\delta} a_\alpha [a_\gamma, a_\beta] a_\delta$ and $R - D^2 = \sum \Gamma_{\alpha\beta} \Gamma_{\gamma\delta} a_\alpha [a_\gamma a_\delta, a_\beta]$.

Remark 1 (Interpretation). • *Commuting case.* If all O_α commute, then $E_1 = 0$, and in fact $\Phi = e^{\delta t \mathcal{L}}$ exactly, because the Gaussian characteristic function is exact. This includes $N_c = 1$.

- *Coherent part.* The $\mu\Gamma$ term is a Hamiltonian correction. Since $[a_\alpha, [a_\beta, M]] = \text{ad}_{[O_\alpha, [O_\beta, H_\mu]]}$ with $H_\mu = \sum \mu_\alpha O_\alpha$, it is $\frac{i\delta t}{12} \sum \Gamma_{\alpha\beta} \text{ad}_{[O_\alpha, [O_\beta, H_\mu]]}$ in the generator, i.e. a colored-noise renormalization $\delta H \propto \frac{\delta t}{12} \sum \Gamma_{\alpha\beta} [O_\alpha, [O_\beta, H_\mu]]$.
- *Noise part.* The $\Gamma\Gamma$ term is a non-Lindbladian, purely noise-induced defect. It is nonzero whenever the noise channels do not commute.
- *Accumulated error.* Over $T = n\delta t$ the averaged error is $\simeq T \delta t \|E_1\|$, so the joint kick is a weak first-order integrator of $\mathcal{L}$. Its size scales as

$$\|E_1\| = \mathcal{O}(\Gamma \mu \|O\|^3 + \Gamma^2 \|O\|^4),$$

with only nested commutators contributing.

3 First-order splitting

3.1 Ordering-defect lemma

Expand both exponentials in words $a_{\alpha_1} \cdots a_{\alpha_n}$. Fix a content, i.e. multiplicities $\{n_\alpha\}$ with $\sum n_\alpha = n$. The joint exponential gives $(1/n!)$ times the sum over all $n!/\prod n_\alpha!$ distinct arrangements. The split product gives only the sorted word, with weight $1/\prod n_\alpha!$. For a symmetric coefficient tensor c (here a moment $\mathbb{E}[X^{\otimes n}]$), define the *ordering defect*

$$P_n[c] = \sum_{\alpha_1 \leq \cdots \leq \alpha_n} \frac{c_{\alpha_1 \cdots \alpha_n}}{\prod_\alpha n_\alpha!} a_{\alpha_1} \cdots a_{\alpha_n} - \frac{1}{n!} \sum_{\alpha_1, \dots, \alpha_n} c_{\alpha_1 \cdots \alpha_n} a_{\alpha_1} \cdots a_{\alpha_n}. \quad (11)$$

Then, exactly,

$$\Phi_{\text{split}} - \Phi = \sum_{n \geq 2} i^n P_n[\mathbb{E}X^{\otimes n}], \quad (12)$$

because $P_1 \equiv 0$ and P_n vanishes on any word with a single distinct index. For $n = 2$,

$$P_2[c] = \frac{1}{2} \sum_{\alpha < \beta} c_{\alpha\beta} [a_\alpha, a_\beta]. \quad (13)$$

3.2 Leading order: an $\mathcal{O}(\delta t)$ coherent bias

Proposition 2.

$$\Phi_{\text{split}} - \Phi = \delta t S_1 + \delta t^2 S_2 + \mathcal{O}(\delta t^3), \quad (14)$$

where

$$S_1 = -\frac{1}{2} \sum_{\alpha < \beta} \Gamma_{\alpha\beta} [a_\alpha, a_\beta] = -\frac{1}{2} \sum_{\alpha < \beta} \Gamma_{\alpha\beta} \text{ad}_{[O_\alpha, O_\beta]}, \quad (15)$$

$$S_2 = -P_2[\mu\mu] - i P_3[S] + P_4[T] = -\frac{1}{2} \sum_{\alpha < \beta} \mu_\alpha \mu_\beta [a_\alpha, a_\beta] - i P_3[S] + P_4[T]. \quad (16)$$

Consequently,

$$\Phi_{\text{split}} = e^{\delta t \mathcal{L}} + \delta t S_1 + \delta t^2 (E_1 + S_2) + \mathcal{O}(\delta t^3). \quad (17)$$

Proof. The $n = 2$ term $i^2 P_2[\Gamma \delta t + \mu\mu \delta t^2]$ gives S_1 and the first part of S_2 . The $n = 3$ term gives $i^3 \delta t^2 P_3[S]$, and the $n = 4$ term gives $\delta t^2 P_4[T]$. Combine with Proposition 1. $\square$

Remark 2 (Physics of S_1). Write $[O_\alpha, O_\beta] = i C_{\alpha\beta}$ with $C_{\alpha\beta}$ Hermitian. Then

$$\delta t S_1 = -i \delta t \text{ad}_{H_{\text{shift}}}, \quad H_{\text{shift}} = \frac{i}{2} \sum_{\alpha < \beta} \Gamma_{\alpha\beta} [O_\alpha, O_\beta] = -\frac{1}{2} \sum_{\alpha < \beta} \Gamma_{\alpha\beta} C_{\alpha\beta}. \quad (18)$$

- *Origin.* The ordered product produces $X_\alpha X_\beta a_\alpha a_\beta$ for $\alpha < \beta$. The dissipator D requires the symmetrized form $\frac{1}{2} \{a_\alpha, a_\beta\}$. The mismatch is a commutator, which acts as a Hamiltonian. This is an Itô/Stratonovich-type ordering bias, and it is $\mathcal{O}(\delta t)$ per step.
- *Consequence.* The averaged split circuit converges to $\mathcal{L} + S_1$, not to $\mathcal{L}$. At fixed total time the error is $\simeq T \|S_1\|$ and does *not* vanish as $\delta t \rightarrow 0$.

Remark 3 (When S_1 vanishes, and fixes). 1. $S_1 = 0$ iff $\sum_{\alpha < \beta} \Gamma_{\alpha\beta} [O_\alpha, O_\beta] \propto \mathbb{K}$. This holds for independent noise (Γ diagonal), or when correlations only connect commuting terms.

2. Reversing the order of the factors flips the sign of S_1 and of the P_2 part of S_2 .
3. The palindromic (Strang) product with the same sample,

$$\mathbb{E}\left[\prod_{\alpha=1}^{N_c} e^{\frac{i}{2}X_\alpha a_\alpha} \prod_{\alpha=N_c}^1 e^{\frac{i}{2}X_\alpha a_\alpha}\right],$$

cancels S_1 . Its logarithm is odd in X , so its second-order part is the symmetrized $\frac{1}{2} \sum X_\alpha X_\beta a_\alpha a_\beta$, and the remaining error is $\mathcal{O}(\delta t^2)$ per step.

4. Diagonalize first: with $\Gamma = BB^T$, kick along independent Gaussian combinations, or keep correlated non-commuting terms inside one joint exponential.
5. If $[O_\alpha, O_\beta] \in \text{span}\{O_\gamma\}$, the bias can be precompensated by shifting μ .

3.3 Diagonal covariance: the split channel is an exact Lindblad splitting

Proposition 3. *If $\Gamma = \text{diag}(\gamma_\alpha)$, the X_α are independent. Each factor then averages exactly to*

$$\mathbb{E}[e^{iX_\alpha a_\alpha}] = e^{\delta t \mathcal{L}_\alpha}, \quad \mathcal{L}_\alpha = i\mu_\alpha a_\alpha - \frac{1}{2}\gamma_\alpha a_\alpha^2,$$

so

$$\Phi_{\text{split}} = \prod_{\alpha} e^{\delta t \mathcal{L}_\alpha}, \quad \Phi_{\text{split}} - e^{\delta t \mathcal{L}} = \frac{\delta t^2}{2} \sum_{\alpha < \beta} [\mathcal{L}_\alpha, \mathcal{L}_\beta] + \mathcal{O}(\delta t^3). \quad (19)$$

This is the ordinary Lie–Trotter error of $\mathcal{L} = \sum_{\alpha} \mathcal{L}_\alpha$: weak order 1, with no stochastic remainder. Note the contrast with the joint channel: for diagonal Γ it still carries the $\Gamma\Gamma$ defect $\frac{1}{24}(Q + R - 2D^2)$. Splitting independent channels removes that defect and replaces it with deterministic Lindblad-splitting commutators. Higher-order Lindblad splittings (Strang, Suzuki) of the $\mathcal{L}_\alpha$ then apply directly.

4 Numerical verification

The test system is two qubits with three mutually non-commuting terms,

$$O_1 = X \otimes \mathbb{K} + 0.3 Z \otimes Z, \quad O_2 = Y \otimes X + 0.5 \mathbb{K} \otimes Z, \quad O_3 = Z \otimes \mathbb{K} + 0.4 X \otimes Y,$$

with random μ and $\Gamma = BB^T/2$ drawn from a Gaussian B (seed 1). Ensemble averages are computed with tensor-product Gauss–Hermite quadrature: 24 nodes per dimension, which is exact for polynomials of degree ≤ 47 . Norms are spectral norms of the 16×16 superoperators. Each column of Table 1 is the relative residual after subtracting the predicted term. It should scale as δt , and it does.

Table 2 gives absolute one-step errors for correlated Γ . The split channel errs at $\mathcal{O}(\delta t)$, while the joint and Strang channels err at $\mathcal{O}(\delta t^2)$.

5 Splitting out the coherent part

Separate the mean from the fluctuation, and apply the coherent part as symmetric half steps around a centred noise kick:

$$\Phi_{\text{cs}} = e^{i\delta t M/2} \mathbb{E}[e^{iY \cdot a}] e^{i\delta t M/2}, \quad Y \sim \mathcal{N}(0, \Gamma \delta t). \quad (20)$$

Residual (relative)	$\delta t = 0.02$	0.01	0.005
$\ \Phi - e^{\delta t \mathcal{L}} - \delta t^2 E_1\ /\ \delta t^2 E_1\ $	5.3×10^{-2}	2.7×10^{-2}	1.4×10^{-2}
$\ \Phi_{\text{split}} - \Phi - \delta t S_1\ /\ \delta t S_1\ $	1.6×10^{-1}	8.3×10^{-2}	4.2×10^{-2}
$\ \Phi_{\text{split}} - \Phi - \delta t S_1 - \delta t^2 S_2\ /\ \delta t^2 S_2\ $	–	3.2×10^{-2}	1.6×10^{-2}
diagonal Γ : $\ \Phi_{\text{split}} - e^{\delta t \mathcal{L}} - \frac{\delta t^2}{2} \sum [\mathcal{L}_\alpha, \mathcal{L}_\beta]\ /\ \cdot\ $	–	3.2×10^{-2}	1.6×10^{-2}

Table 1: Relative residuals after subtracting the predicted leading terms. Halving δt halves each residual, confirming the next order.

One-step error	$\delta t = 0.02$	0.01	0.005
joint, $\ \Phi - e^{\delta t \mathcal{L}}\ $	6.0×10^{-4}	1.5×10^{-4}	3.9×10^{-5}
split, $\ \Phi_{\text{split}} - e^{\delta t \mathcal{L}}\ $	2.8×10^{-2}	1.4×10^{-2}	7.0×10^{-3}
Strang (same sample), $\ \Phi_{\text{Strang}} - e^{\delta t \mathcal{L}}\ $	1.4×10^{-3}	3.7×10^{-4}	9.5×10^{-5}

Table 2: Absolute one-step errors for correlated Γ .

Proposition 4.

$$\Phi_{\text{cs}} = e^{\delta t \mathcal{L}} + \delta t^2 \Delta + \mathcal{O}(\delta t^3), \quad \Delta = \frac{1}{24}(Q + R - 2D^2). \quad (21)$$

Proof. Applying Proposition 1 with $\mu = 0$ gives $\mathbb{E}[e^{iY \cdot a}] = e^{-\delta t D/2} + \delta t^2 \Delta + \mathcal{O}(\delta t^3)$. The sandwich is then a Strang splitting of $\mathcal{L} = A + B$, with $A = iM$ and $B = -D/2$. Its error, $\delta t^3(\frac{1}{12}[B, [B, A]] - \frac{1}{24}[A, [A, B]])$, enters only at third order. $\square$

Comparison with the joint kick. The coherent-noise cross term $\frac{i}{12} \sum \Gamma_{\alpha\beta}[a_\alpha, [a_\beta, M]]$ of E_1 is removed, and only the noise-only defect Δ remains.

- *Size of the gain.* The cross term scales as $\Gamma\mu$ and Δ scales as Γ^2 , so the gain is $\sim \mu/\Gamma$. In the model of Table 1 the one-step error drops by $1.6\times$ at $|\mu| \sim \Gamma$ and by $7\text{--}8\times$ when μ is scaled up by 10.
- *Cost.* Adjacent coherent half steps merge across time steps, so the depth equals that of a Lie ordering.
- *One-sided ordering.* The ordering $e^{i\delta t M} \mathbb{E}[e^{iY \cdot a}]$ reinstates an $\mathcal{O}(\delta t^2)$ cross term, $-\frac{i}{4}[M, D]$.
- *Splitting the kick afterwards.* If the centred kick is split sequentially, the bias S_1 of Proposition 2 survives, because it does not involve μ .

6 Zero-field NMR benchmark

Model. We consider rigid molecules with intramolecular dipolar relaxation under isotropic rotational diffusion, with $\tau_c = 2$ ps. The coherent part is the isotropic J network, $H_J = \sum_{i < j} 2\pi J_{ij} \mathbf{I}_i \cdot \mathbf{I}_j$. For each spin pair p the dipolar coupling is $b_p = (\mu_0/4\pi)\gamma_i\gamma_j\hbar/r_p^3$. The noise operators are the five real rank-2 components $A_m^{(p)} = \sum_{ab} e_{ab}^{(m)} I_{i,a} I_{j,b}$, where $\{e^{(m)}\}$ is a Frobenius-orthonormal basis of traceless symmetric 3×3 matrices. Isotropic averaging and the white-noise limit give

$$\Gamma_{(pm),(p'm')} = G_{pp'} \delta_{mm'}, \quad G_{pp'} = 2\tau_c b_p b_{p'} \frac{6}{5} P_2(\hat{n}_p \cdot \hat{n}_{p'}), \quad (22)$$

so dipolar cross-correlations make Γ off-diagonal between pairs. Three regime-specific facts:

- At zero field all spectral densities are sampled at $\omega \sim 2\pi J$. Since $J\tau_c \sim 10^{-9}$, the white-noise limit is essentially exact.
- There is no secular truncation and no Zeeman frame.
- CSA is absent because it scales with B_0 . Spin-rotation and paramagnetic random fields are omitted here.

The molecules are ^{13}C -formic acid (CH , $J = 222 \text{ Hz}$), a $^{13}\text{CH}_2$ group ($J_{\text{CH}} = 140 \text{ Hz}$, $J_{\text{HH}} = -12 \text{ Hz}$, tetrahedral, $r_{\text{CH}} = 1.09 \text{ \AA}$), and a $^{13}\text{CH}_3$ group ($J_{\text{CH}} = 136 \text{ Hz}$). For these, $\|\mathcal{L}_H\|/\|D\| \approx 1.1 \times 10^4$, 3.5×10^3 and 2.5×10^3 , so the coherent part dominates by 3–4 orders of magnitude.

Schemes. In all schemes the coherent block $e^{-iH_J\delta t}$ (or its half) is applied exactly, so the table isolates the noise-induced error.

1. *Simplest palindromic*: the sequence $A_1/2, \dots, A_K/2, H_J, A_K/2, \dots, A_1/2$ over the physical term list, with one sample per step. The coherent block sits at the centre.
- 1'. As 1, but with the coherent block outermost.
2. *Coherent–incoherent*: $e^{-iH_J\delta t/2} \mathbb{E}[e^{-iY \cdot A}] e^{-iH_J\delta t/2}$, with a joint centred kick (Section 5).
3. *Noise eigenbasis*: write $\Gamma = \sum_k \lambda_k v_k v_k^T$ and $A'_k = \sum_\alpha v_{k\alpha} A_\alpha$. Then apply coherent halves around a Strang sequence $A'_1/2, \dots, A'_{K'}/2, \dots, A'_1/2$ of independent kicks. *The two half-kicks of each mode use independent draws of variance $\lambda_k \delta t/2$, so each averages exactly to $e^{\delta t \mathcal{L}_k/2}$ and the averaged step is a deterministic Strang splitting of $\mathcal{L}$.*
- 3'. As 3, but with the same sample in both halves.

We also give two reference schemes: the joint unsplit kick, and the Lie (sequential) split in the physical basis.

Method. The averaged channels of schemes 1, 1', 2 and 3' are computed with a fully symmetric degree-5 Gaussian cubature ($2K^2 + 1$ nodes). This is exact for every term of $\mathcal{O}(\delta t^2)$ and for all $\mu^2\Gamma$ and $\mu\Gamma^2$ terms at $\mathcal{O}(\delta t^3)$. As a check, it reproduces $\delta t^2 E_1$ for the joint kick with a residual $\propto \delta t$. Scheme 3 is computed exactly. We report $\|\Phi^n - e^{T\mathcal{L}}\|$ (superoperator spectral norm) at $T = 1 \text{ s}$, with $2\pi J_{\text{max}}\delta t = 0.1$. The observed global order p is obtained by halving δt .

Observations.

1. *Coherent placement is the dominant effect.* With ≥ 3 spins, putting the coherent block inside the palindrome (scheme 1), or inside the kick (joint), leaves the $\Gamma\mu$ cross term. This is $\sim 3 \times 10^3 \times$ worse than any coherent-outermost scheme, in line with $\mu/\Gamma \sim 10^3$.
2. *Two-spin exception.* For an isolated pair, $[\mathbf{I} \cdot \mathbf{S}, A_m^{(p)}] = 0$: a rank-2 tensor of the pair cannot connect the singlet to any state. Coherent and dissipative parts therefore commute exactly, all schemes agree, and only noise–noise defects remain.
3. *Local versus global error.* Schemes 2, 1', 3' and 3 have the same local error, set by the $\mu^2\Gamma\delta t^3$ Strang term. That term has the form $[\mathcal{L}_H, \cdot]$, is averaged out by the J precession, and does not accumulate: the global error of scheme 3 is $\sim 10^{-4}$ of the linear sum of its local errors. What accumulates in 2, 1' and 3' is the noise-only defect $\Delta \sim \Gamma^2$, giving global order 1.

Scheme	CH ($K = K' = 5$)		CH ₂ ($K = K' = 15$)		CH ₃ ($K = 30, K' = 25$)	
	error	p	error	p	error	p
1 palindromic, coherent centre	2.9×10^{-8}	1	1.4×10^{-3}	1	3.2×10^{-3}	1
1' palindromic, coherent outermost	2.9×10^{-8}	1	4.5×10^{-7}	1	8.5×10^{-7}	1
2 coh. joint centred kick coh.	3.8×10^{-8}	1	5.7×10^{-7}	1	1.4×10^{-6}	1
3 coh. eigenmode Strang (indep. halves) coh.	$<10^{-11}$	–	1.3×10^{-9}	2	5.5×10^{-9}	2
3' as 3, same sample in both halves	2.9×10^{-8}	1	5.4×10^{-7}	1	1.0×10^{-6}	1
ref. joint unsplit	3.8×10^{-8}	1	9.2×10^{-4}	1	2.2×10^{-3}	1
ref. Lie, physical basis	8.9×10^{-9}	1	7.0×10^{-7}	0	8.3×10^{-3}	0

Table 3: Global error at $T = 1$ s for zero-field ^{13}C – ^1H systems with $2\pi J_{\max}\delta t = 0.1$ ($\delta t = 72, 114$ and $117 \mu\text{s}$). K is the number of physical noise operators and $K' = 5 \text{rank } G$ the number of eigenmodes. For CH, schemes 1' and 3' coincide. The order $p = 0$ marks an $\mathcal{O}(T)$ bias that does not decrease with δt .

4. *Why scheme 3 wins.* Scheme 3 removes Δ , reaching global order 2 and an error 250 – $450\times$ below scheme 2. The gain comes from *independent* draws for the two half-kicks. The eigenbasis matters because it is what makes independent draws legitimate. With the same sample (3'), the eigenbasis buys nothing over 1'.
5. *Sequential split in the physical basis.* The Lie split is biased ($p = 0$). For CH₃ the bias reaches 8×10^{-3} . Only the part of H_{shift} that commutes with H_J accumulates, which is why the bias is much smaller for CH₂.
6. *Cost.* The exponentials per step after merging are: 1: $2K$; 1': $2K$; 2: two, but the kick $e^{-iY \cdot A}$ is a dense K -term exponential; 3: $2K'$. In scheme 3 the mode operators A'_k are collective, each a sum over pairs of one tensor component. Since $\text{rank } G \leq 5$, CH₃ needs 25 modes instead of 30.
7. *Coherent Trotterization.* If H_J is itself Trotterized, its deterministic error ($\propto (J\delta t)^3$ per step) adds equally to all schemes and is controlled separately.

7 Summary

- **Joint kick.** $\Phi = e^{\delta t \mathcal{L}} + \delta t^2 E_1$. There are two defects: a coherent colored-noise shift $\frac{i}{12} \sum \Gamma_{\alpha\beta} [a_\alpha, [a_\beta, M]]$, and a quartic noise defect $\frac{1}{24}(Q + R - 2D^2)$. The global error is $\mathcal{O}(T\delta t)$. It is exact for commuting O_α .
- **Sequential split.** $\Phi_{\text{split}} = e^{\delta t \mathcal{L}} + \delta t S_1 + \mathcal{O}(\delta t^2)$, with $S_1 = -i \text{ad}_{H_{\text{shift}}}$ and $H_{\text{shift}} = \frac{i}{2} \sum_{\alpha < \beta} \Gamma_{\alpha\beta} [O_\alpha, O_\beta]$. This is an $\mathcal{O}(T)$ bias that does not vanish as $\delta t \rightarrow 0$ whenever correlated noise sits on non-commuting terms.
- **Diagonal Γ .** Splitting is exact Lindblad Lie–Trotter: $\frac{\delta t^2}{2} \sum_{\alpha < \beta} [\mathcal{L}_\alpha, \mathcal{L}_\beta]$.
- **Coherent split out first.** $\Phi_{\text{cs}} = e^{\delta t \mathcal{L}} + \delta t^2 \Delta$. This removes the $\Gamma\mu$ cross term, a gain of $\sim \mu/\Gamma$, i.e. $\sim 10^3$ in zero-field NMR. It is exact for an isolated spin pair.
- **Prescription.** Use coherent halves outermost, then Strang over the noise eigenmodes with independent half-kick draws. This gives global order 2 and removes the Γ^2 defect Δ (Table 3). Never split correlated noise on non-commuting terms sequentially.